

Lab # 09

Introduction to Functions in C++

Objective

To understand the concept of functions in C++ and learn how to define and call functions in a program.

Apparatus / Software Required

- Computer
- C++ Compiler (Code::Blocks / Dev C++ / Visual Studio / Online Compiler)

Theory

A **function** in C++ is a block of code that performs a specific task. Functions help to:

- Organize code properly
- Avoid repetition
- Make programs easier to read and maintain

Types of Functions:

1. **Built-in functions** (e.g., cout, cin)
2. **User-defined functions** (created by the programmer)

Basic Structure of a Function:

```
return_type function_name(parameters)
{
    // body of function}
```

Example Program

```
#include <iostream>
using namespace std;

// Function declaration
// yahan hum function ka sirf naam aur return type bata rahe hain
void greet();

int main() {
    greet(); // function call kar rahe hain jo neeche define hai
    return 0; // program successfully khatam ho gaya
}

// Function definition
// yahan function ka actual kaam likha gaya hai
void greet() {
    cout << "Hello! Welcome to C++ Functions Lab." << endl; // screen par message print karta hai
}
```

Working

- The function greet() is declared before main().
- Inside main(), the function is called.
- The function prints a greeting message.

Output

Hello! Welcome to C++ Functions Lab.

Conclusion

In this lab, we learned how to create and use functions in C++. Functions make programs easier to manage and improve code reusability.

Lab 09 Tasks

Task 01:

Function Introduction and Declaration

- Declare a function called "greet" that takes no parameters and returns nothing.
- Inside the function, display a greeting message of your choice.
- Call the function from the main program to execute the greeting.

Code:

```
#include <iostream>
using namespace std;

// Function declaration
void greet();

int main() {
    greet();    // Function call
    return 0;
}

// Function definition
void greet() {
    cout << "Welcome to C++ Programming Lab!" << endl;
}
```

Task 02:

Basic Function Declaration and Call

- Declare a function called "sayHello" that takes no parameters and returns nothing.
- Inside the function, display the message "Hello, World!".
- Call the function from the main program to execute the greeting.

Code:

```
#include <iostream>
using namespace std;

// Function declaration
// yahan hum function ka naam aur return type define kar rahe hain
void sayHello();

int main() {
    sayHello();    // yahan function ko call kiya ja raha hai
    return 0;      // program yahan successfully end ho jata hai
}

// Function definition
// yahan function ka actual kaam likha gaya hai
void sayHello() {
    cout << "Hello, World!" << endl;    // yeh line screen par message print karti hai
}
```

Task 02:

Write a C++ program that:

- Takes 10 integer values as input from the user and stores them in a 1D array.
 - Creates a function named calculateSum that:
 - Accepts the array and its size as parameters Returns the sum of all elements In the main() function:
 - Call the function Calculate and display the average of the array elements
- Constraints:
- Use only one-dimensional array Use function for sum calculation

Code:

```
#include <iostream>
using namespace std;

// Function to calculate sum
// yeh function array ke tamam elements ka sum calculate karta hai
int calculateSum(int arr[], int size) {
    int sum = 0; // yahan sum ko 0 se initialize kiya gaya hai

    for(int i = 0; i < size; i++) {
        sum = sum + arr[i]; // har element ko sum mein add kiya ja raha hai
    }

    return sum; // final sum wapas return kiya ja raha hai
}

int main() {
    int arr[10]; // 10 integers store karne ke liye array banaya gaya hai
    int sum; // sum store karne ke liye variable

    // Input from user
    // user se 10 numbers input lene ke liye
    cout << "Enter 10 integers: " << endl;
    for(int i = 0; i < 10; i++) {
        cin >> arr[i]; // har index par user se value li ja rahi hai
    }

    // Function call
    // yahan function ko call karke sum hasil kiya ja raha hai
    sum = calculateSum(arr, 10);

    // Calculate average
    // yahan average calculate kiya ja raha hai
    float average = sum / 10.0; // 10.0 use kiya taake float result mile
```

```
// Output results
// yahan final results display kiye ja rahe hain
cout << "Sum = " << sum << endl;
cout << "Average = " << average << endl;

return 0; // program successfully end hota hai
}
```